(Please write your Exam Roll No.) Roll No.....

Midterm Examination

8th Semester [B.TECH] **April-2024**

Time: 1: 30 hrs **Max Marks: 30**

Paper Code: - ETCS412 **Subject: -Object Oriented Software Engineering**

Note: Attempt any three questions including Q.No. 1 which is compulsory.

Q.1 (a)	Define Object Oriented Software Engineering.	**2**	**CO1**
(b)	Differentiate between ER diagram and use case diagram.	**2**	**CO1**
(c)	Explain the importance of modeling.	**2**	**CO2**
(d)	Differentiate between data abstraction and encapsulation.	**2**	**CO1**
(e)	What do you mean by refactoring?	**2**	**CO2**
Q2 (a)	What is OOA and OOD? What is the relationship between OOA, software requirement and OOD?	**5**	**CO1**
(b)	What are visitor patterns? How are they implemented? What are the rules followed by visitor patterns?	**5**	**CO2**
Q3.(a)	What do you mean by system development? Also discuss different models of system development.	**5**	**CO1**
(b)	What are object oriented methodologies. What is the importance and principles of modeling?	**5**	**CO2**
Q4.(a)	What are the different types of requirement elicitation and give some techniques of requirement elicitation? What are the various challenges faced during requirement elicitation	**5**	**CO1**
(b)	Define the object oriented software engineering concept of association, aggregation and generalization/specialisation with examples.	**5**	**CO1**

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END TERM EXAMINATION

FIFTH SEMESTER [B.TECH] JANUARY 2024

Paper Code: CIC-309 ***Subject: Software Engineering***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit. Use of simple Calculator is allowed.

Q1 Answer **any five** questions:- **(5x3=15)**

(a) Briefly explain the concept of Software Crisis and how it differs from common Software Myths. Provide real-world examples.

(b) Distinguish software re-engineering and reverse engineering?

(c) Calculate and interpret Halstead's effort for a program where the number of distinct operators is n_1=15, the number of distinct operands is n_2=20, the total number of operators isN_1=120, and the total number of operands is N_2=80.

(d) Discuss the implications of high coupling on software maintainability, with examples from industry practices.

(e) Discuss the role of regression testing in Agile methodologies.

(f) Draw the Context level 0 and level 1 DFD for the Safe home Software?

(g) Assess the effectiveness of data dictionaries in managing large-scale software projects, citing real-world applications.

(h) Develop a set of test cases to perform boundary value analysis for the following input fields in an online registration form:

(i) Age: The form requires students to enter their age, which must be between 18 and 30 years inclusive.

(ii) Number of Completed High School Subjects: The form also asks for the number of completed high school subjects, which should be a minimum of 5 and a maximum of 12.

Ensure your test cases cover the boundary conditions for these input fields.

UNIT-I

Q2 (a) Compare and contrast the Waterfall and evolutionary Models of SDLC. Focus on key stages, flexibility, risk management, and suitability for different types of projects. Provide examples. **(7.5)**

(b) Explain the importance and steps of requirement elicitation in software engineering. Discuss the FAST and Use Case approaches, highlighting their strengths and limitations. **(7.5)**

Q3 (a) Describe the key characteristics and structure of an effective Software Requirement Specification (SRS) document. Use IEEE standards as a reference. **(7.5)**

(b) Discuss the differences and similarities between Functional and Non-Functional requirements with examples. How do these impact the software development process? **(7.5)**

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UNIT-II

Q4 (a) Explain the COCOMO cost estimation model. Perform a basic COCOMO calculation for a project with a given size (50 KLOC). Using the Intermediate COCOMO model, calculate the effort in person-months. Assume the project has 'HIGH' complexity [CMPLX=High=1.15]. **(5)**

(b) Describe the concepts of cohesion and coupling in software design. Provide examples of each and discuss their impact on software maintainability and flexibility. **(5)**

(c) Write a short note on live variables and module weakness. **(5)**

Q5 (a) Discuss the Putnam Resource Allocation Model, including its application and limitations. Perform a calculation using given project parameters. **(5)**

(b) Compare and contrast Object-Oriented Design (OOD) and Function-Oriented Design (FOD) in terms of their methodology, advantages, and disadvantages. **(5)**

(c) Differentiate between Function Point and LOC software Metrics. Why Function points metrics are preferred over Lines of Code Metrics. **(5)**

UNIT-III

Q6 (a) Explain Halstead's Software Science metrics. Write equations to calculate the Effort and Time for a given program using Halstead's methos. **(7.5)**

(b) Describe the concept of Software Reliability. Compare the Basic Model and Logarithmic Poisson Model in terms of their approach and applicability. **(7.5)**

Q7 (a) Discuss the importance of software metrics in software engineering. Explain Token Count and Information Flow metrics with examples. **(7.5)**

(b) Explain the CMM (Capability Maturity Model) and ISO 9001 standards. Discuss how they contribute to software quality assurance. Give example of three CMM Level 3 companies based in India. **(7.5)**

UNIT-IV

Q8 (a) Describe the testing process in software engineering, focusing on the importance of functional and structural testing. Provide examples of test cases for each. **(5)**

(b) Create Control Flow Graph for a module that checks if sum of 2 numbers is positive, negative or neutral. Also find cyclomatic complexity number for this module and number of independent paths in the module. **(5)**

(c) Explain the concept and process of Software Maintenance. Discuss the Quick-fix and Boehm's models of maintenance with their merits and demerits. **(5)**

Q9 (a) Define and differentiate between Verification and Validation in software testing. Provide examples to illustrate your points. **(5)**

(b) Discuss the importance and methods of Software Configuration Management in the context of software maintenance. **(5)**

(c) Explain the importance of acceptance testing in software development. Compare and contrast alpha and beta testing. **(5)**

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MID TERM EXAMINATION

EIGHTHSEMESTER [B. TECH.] APRIL2022

Subject: Object Oriented Software Engineering

Paper Code: ETCS412

Time: 1Hour 30 Min. **Maximum Marks: 30**

Note: Attempt any Three questions including Q. No. 1 which is compulsory.

Q1(a). Explain OOA and OOD. (2)

(b). Explain Include and Extend relationship in use case diagram. (2)

(c). Draw the OOSD life cycle model by Jacobson. (2)

(d). What is good object model. (2)

(e). Distinguish between functional and non-functional requirements. (2)

Q2. What is requirement elicitation and explain its any two techniques? (10)

Q3. (a) Compare ER diagram and Use Case diagram. (5)

(b). Explain Inheritance and its type with example. (5)

Q4. Explain Use case model and Draw a Use case diagram of college examination system. (10)

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Mid–Term Examination

VIII Semester [B.Tech] March, 2022

Paper Code: ETCS-412 Sub: OOSE

Time: 1.5 Hrs Max. Marks: 30

Note: Q.No.1 is compulsory and attempt any two more questions from the remaining.

Q1. (2.5*5)

a) What for refactoring is used?

b) Explain Encapsulation concept of Object Oriented.

c) What do you mean by requirement elicitation?

d) What are functional and non-functional requirements?

Q2. a) Explain Anti-patterns and visitor-patterns in detail. (5)

b) Explain inheritance with the help of an example. (5)

Q-3 a) What is the difference between Analysis model and design model (5)

b) What are the advantages and principles of modelling (5)

Q4. a) Draw Use Case diagram of Railway Management System. Depict the notations used. (5)

b) Define the Object-Oriented Software Engineering concepts of association, aggregation and generalization/specialization with examples. (5)